Fashion-MNIST — clustering results report

Interactive view of `results.csv` and figures under `figures/`.

Run Jupyter from the project root (same folder as `main.py`).

```
pip install jupyter ipywidgets
jupyter notebook results_report.ipynb
```

In [1]:
```
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display, Image, Markdown

sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams["figure.dpi"] = 110

ROOT = Path.cwd()
if not (ROOT / "results.csv").exists():
    raise FileNotFoundError(
        "results.csv not found. cd to the project directory or set ROOT t
    )

CSV_PATH = ROOT / "results.csv"
FIG_DIR = ROOT / "figures"

df = pd.read_csv(CSV_PATH)
df.head()
```

Out[1]:

	DimReduction	Clustering	DR_TrainTime_s	Clust_Time_s	N_Cluster
0	Raw	MiniBatchKMeans	0.0	4.985	
1	Raw	DBSCAN	0.0	0.570	
2	Raw	AgglomerativeClustering	0.0	8.226	
3	Raw	GaussianMixture	0.0	21.958	
4	Raw	Birch	0.0	9.774	

Full results table

Higher is better: **Calinski–Harabasz**, **Silhouette**, **Adjusted Rand**.
Lower is better: **Davies–Bouldin**.

In [2]:
```
pd.set_option("display.max_rows", None)
pd.set_option("display.width", 200)
```

```
pd.set_option("display.float_format", lambda x: f"{x:.4f}" if pd.notna(x)
display(df)
```

	DimReduction	Clustering	DR_TrainTime_s	Clust_Time_s	N_Clusters
0	Raw	MiniBatchKMeans	0.0000	4.9850	
1	Raw	DBSCAN	0.0000	0.5700	
2	Raw	AgglomerativeClustering	0.0000	8.2260	
3	Raw	GaussianMixture	0.0000	21.9580	
4	Raw	Birch	0.0000	9.7740	
5	PCA	MiniBatchKMeans	6.1330	1.6980	
6	PCA	DBSCAN	6.1330	0.4430	
7	PCA	AgglomerativeClustering	6.1330	8.7570	
8	PCA	GaussianMixture	6.1330	26.7990	
9	PCA	Birch	6.1330	8.2830	
10	SAE	MiniBatchKMeans	221.2140	1.7110	
11	SAE	DBSCAN	221.2140	0.5280	
12	SAE	AgglomerativeClustering	221.2140	10.4140	
13	SAE	GaussianMixture	221.2140	19.8660	
14	SAE	Birch	221.2140	7.8540	
15	CNN-SAE	MiniBatchKMeans	1156.8490	1.7050	
16	CNN-SAE	DBSCAN	1156.8490	0.5420	
17	CNN-SAE	AgglomerativeClustering	1156.8490	10.5540	
18	CNN-SAE	GaussianMixture	1156.8490	22.2970	
19	CNN-SAE	Birch	1156.8490	9.3380	
20	t-SNE	MiniBatchKMeans	97.1980	1.8920	
21	t-SNE	DBSCAN	97.1980	0.0660	
22	t-SNE	AgglomerativeClustering	97.1980	5.7450	
23	t-SNE	GaussianMixture	97.1980	0.4360	
24	t-SNE	Birch	97.1980	1.1920	
25	UMAP	MiniBatchKMeans	203.2760	2.0250	
26	UMAP	DBSCAN	203.2760	0.4810	
27	UMAP	AgglomerativeClustering	203.2760	9.3300	
28	UMAP	GaussianMixture	203.2760	9.1020	
29	UMAP	Birch	203.2760	0.3180	

Best runs (by metric)

Rows with missing metrics (e.g. single-cluster DBSCAN) are skipped where needed.

In [3]:
```
def best_row(series_col, higher=True):
    s = df[series_col].dropna()
    if s.empty:
        return None
    idx = s.idxmax() if higher else s.idxmin()
    return df.loc[idx]

summary = pd.DataFrame(
    {
        "metric": [
            "Silhouette (↑)",
            "Calinski–Harabasz (↑)",
            "Davies–Bouldin (↓)",
            "Adjusted Rand (↑)",
        ],
        "DimReduction": [
            best_row("Silhouette")["DimReduction"],
            best_row("CalinskiHarabasz")["DimReduction"],
            best_row("DaviesBouldin", higher=False)["DimReduction"],
            best_row("AdjRandIndex")["DimReduction"],
        ],
        "Clustering": [
            best_row("Silhouette")["Clustering"],
            best_row("CalinskiHarabasz")["Clustering"],
            best_row("DaviesBouldin", higher=False)["Clustering"],
            best_row("AdjRandIndex")["Clustering"],
        ],
        "value": [
            best_row("Silhouette")["Silhouette"],
            best_row("CalinskiHarabasz")["CalinskiHarabasz"],
            best_row("DaviesBouldin", higher=False)["DaviesBouldin"],
            best_row("AdjRandIndex")["AdjRandIndex"],
        ],
    }
)
display(summary)
```

	metric	DimReduction	Clustering	value
0	Silhouette (↑)	UMAP	AgglomerativeClustering	0.4893
1	Calinski–Harabasz (↑)	UMAP	AgglomerativeClustering	20009.5703
2	Davies–Bouldin (↓)	UMAP	AgglomerativeClustering	0.7147
3	Adjusted Rand (↑)	UMAP	Birch	0.4704

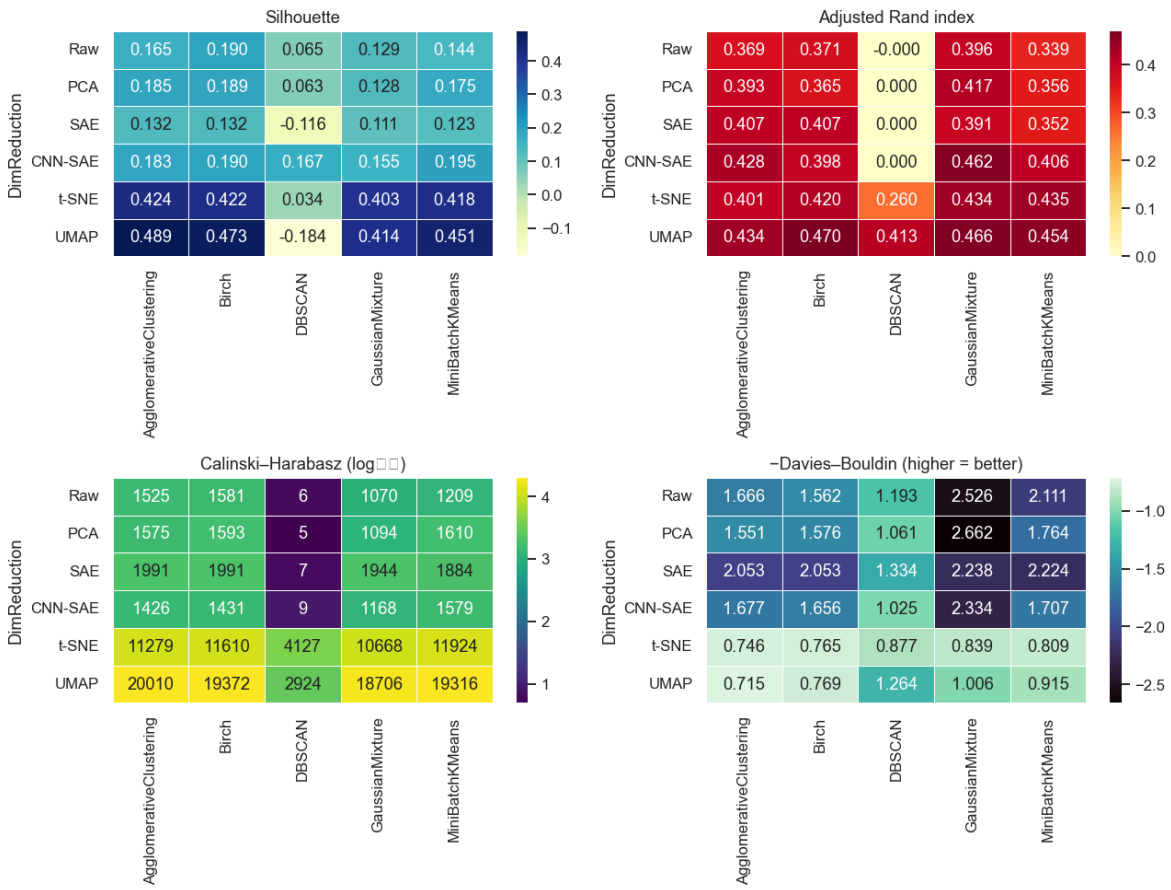
Heatmaps (DR × clustering algorithm)

Davies–Bouldin: colour uses **negative** values so that **darker = better** (same intuition as other panels).

In [4]:

```
def pivot(metric):
    p = df.pivot(index="DimReduction", columns="Clustering", values=metri
    order_dr = [x for x in ["Raw", "PCA", "SAE", "CNN-SAE", "t-SNE", "UMA
    p = p.reindex(order_dr)
    return p

fig, axes = plt.subplots(2, 2, figsize=(12, 9))
metrics = [
    ("Silhouette", "Silhouette", "YlGnBu"),
    ("AdjRandIndex", "Adjusted Rand index", "YlOrRd"),
    ("CalinskiHarabasz", "Calinski-Harabasz (log₁₀)", "viridis"),
    ("DaviesBouldin", "-Davies-Bouldin (higher = better)", "mako"),
]

for ax, (col, title, cmap) in zip(axes.ravel(), metrics):
    p = pivot(col)
    if col == "CalinskiHarabasz":
        plot_p = np.log10(p.clip(lower=1e-12))
        annot = p.map(lambda v: f"{v:.0f}" if pd.notna(v) else "")
    elif col == "DaviesBouldin":
        plot_p = -p
        annot = p.map(lambda v: f"{v:.3f}" if pd.notna(v) else "")
    else:
        plot_p = p
        annot = p.map(lambda v: f"{v:.3f}" if pd.notna(v) else "")
    sns.heatmap(
        plot_p,
        ax=ax,
        cmap=cmap,
        annot=annot,
        fmt="",
        linewidths=0.5,
    )
    ax.set_title(title)
    ax.set_xlabel("")
plt.tight_layout()
plt.show()
```

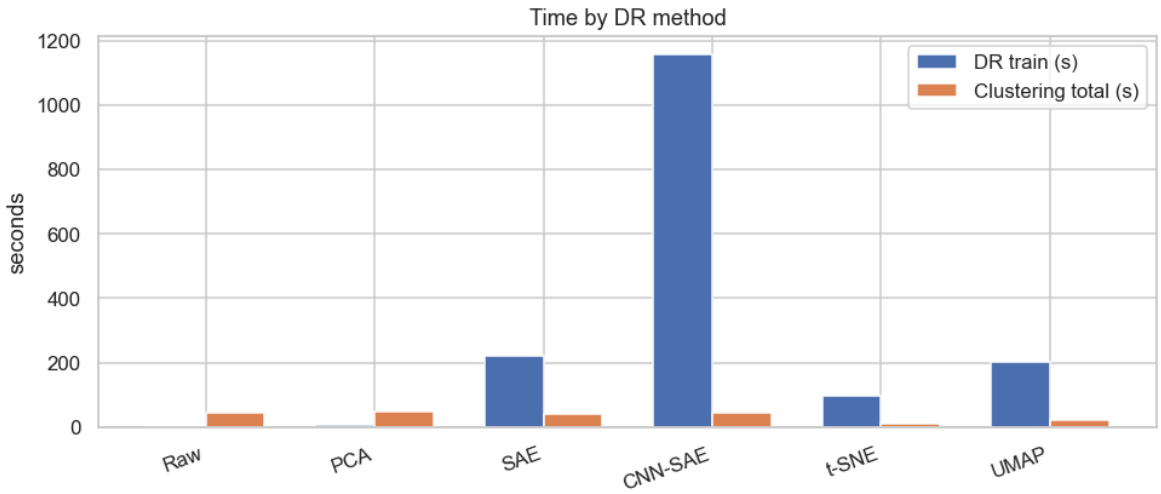
```
/Users/stanimeros/Documents/csc/fashion-mnist-clustering-and-visualizatio
n/.venv/lib/python3.11/site-packages/seaborn/utils.py:61: UserWarning: Gly
ph 8321 (\N{SUBSCRIPT ONE}) missing from font(s) Arial.
  fig.canvas.draw()
/Users/stanimeros/Documents/csc/fashion-mnist-clustering-and-visualizatio
n/.venv/lib/python3.11/site-packages/seaborn/utils.py:61: UserWarning: Gly
ph 8320 (\N{SUBSCRIPT ZERO}) missing from font(s) Arial.
  fig.canvas.draw()
/Users/stanimeros/Documents/csc/fashion-mnist-clustering-and-visualizatio
n/.venv/lib/python3.11/site-packages/IPython/core/pylabtools.py:170: UserW
arning: Glyph 8321 (\N{SUBSCRIPT ONE}) missing from font(s) Arial.
  fig.canvas.print_figure(bytes_io, **kw)
/Users/stanimeros/Documents/csc/fashion-mnist-clustering-and-visualizatio
n/.venv/lib/python3.11/site-packages/IPython/core/pylabtools.py:170: UserW
arning: Glyph 8320 (\N{SUBSCRIPT ZERO}) missing from font(s) Arial.
  fig.canvas.print_figure(bytes_io, **kw)
```



Training / clustering time

Per **dimensionality reduction** block: DR training time is shared across all clustering rows for that DR.

In [5]:
```
dr_order = [x for x in ["Raw", "PCA", "SAE", "CNN-SAE", "t-SNE", "UMAP"]
t_dr = df.groupby("DimReduction", sort=False)["DR_TrainTime_s"].first().r
t_clust = df.groupby("DimReduction", sort=False)["Clust_Time_s"].sum().re

fig, ax = plt.subplots(figsize=(9, 4))
x = np.arange(len(dr_order))
w = 0.35
ax.bar(x - w / 2, t_dr, w, label="DR train (s)")
ax.bar(x + w / 2, t_clust, w, label="Clustering total (s)")
ax.set_xticks(x)
ax.set_xticklabels(dr_order, rotation=20, ha="right")
ax.set_ylabel("seconds")
ax.legend()
ax.set_title("Time by DR method")
plt.tight_layout()
plt.show()

pd.DataFrame({"DR_train_s": t_dr, "clustering_sum_s": t_clust})
```



Out[5]:

DimReduction	DR_train_s	clustering_sum_s
Raw	0.0000	45.5130
PCA	6.1330	45.9800
SAE	221.2140	40.3730
CNN-SAE	1156.8490	44.4360
t-SNE	97.1980	9.3310
UMAP	203.2760	21.2560

Figures

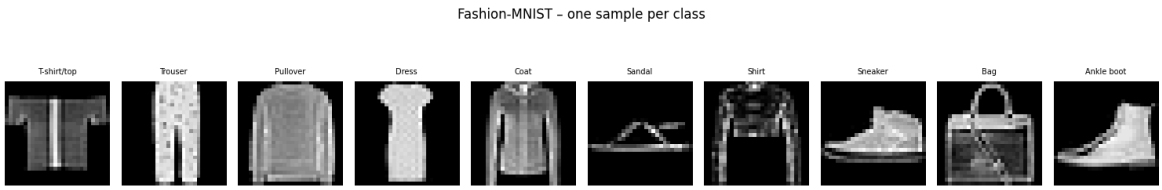
PNG files generated by `main.py` (paths relative to project root).

In [6]:

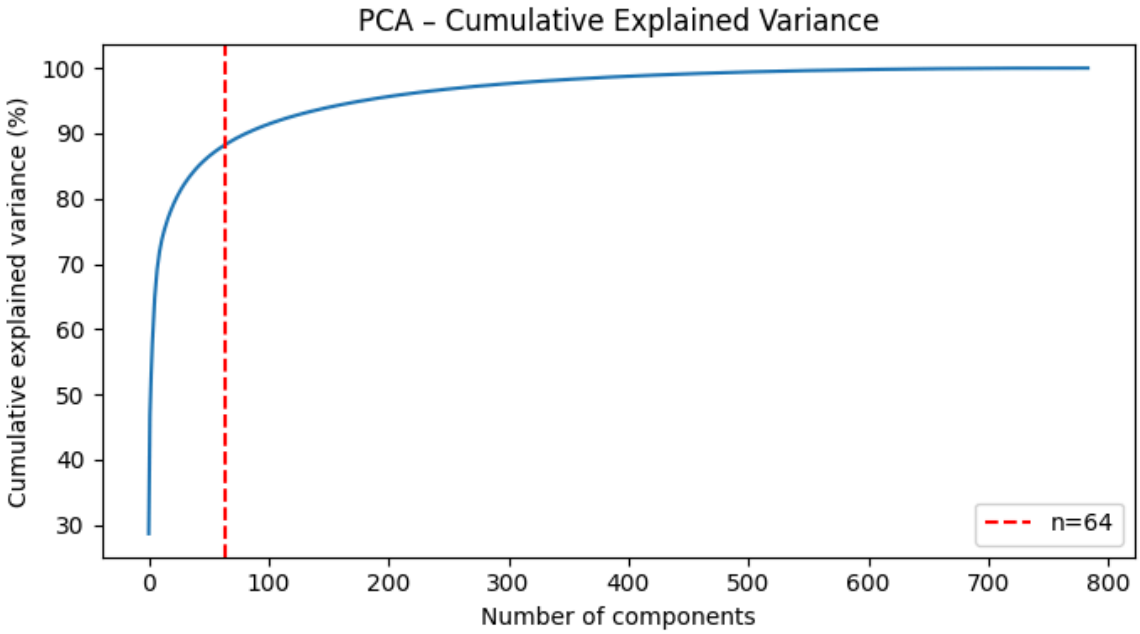
```
FIG_ORDER = [
    "00_sample_images.png",
    "pca_explained_variance.png",
    "sae_reconstructions.png",
    "cnnsae_reconstructions.png",
    "umap_2d.png",
    "scatter_Raw.png",
    "scatter_PCA.png",
    "scatter_SAE.png",
    "scatter_CNN-SAE.png",
    "scatter_t-SNE.png",
    "scatter_UMAP.png",
    "heatmap_silhouette.png",
    "cluster_examples_Raw.png",
    "cluster_examples_PCA.png",
    "cluster_examples_SAE.png",
    "cluster_examples_CNN-SAE.png",
    "cluster_examples_t-SNE.png",
    "cluster_examples_UMAP.png",
]

for name in FIG_ORDER:
    path = FIG_DIR / name
    if not path.is_file():
```

```
        display(Markdown(f"*Missing: `{path}`*"))
        continue
    display(Markdown(f"### `{name}`"))
    display(Image(filename=str(path)))
```

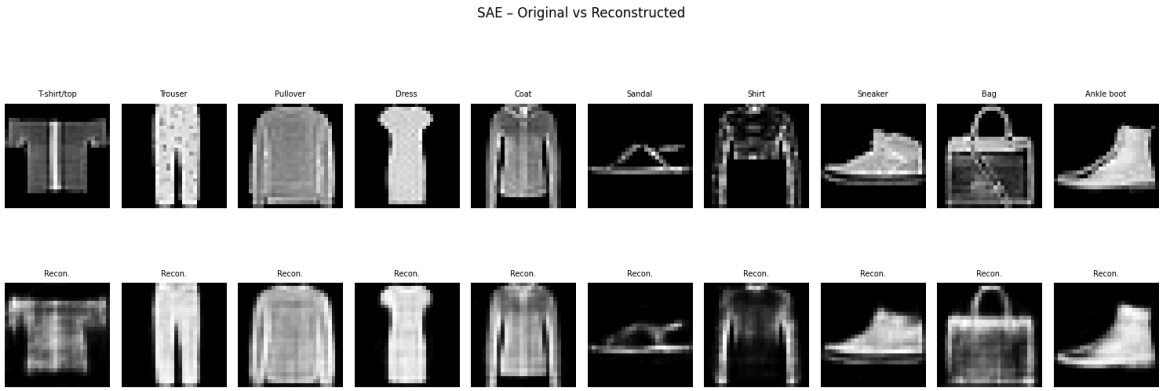
`00_sample_images.png`



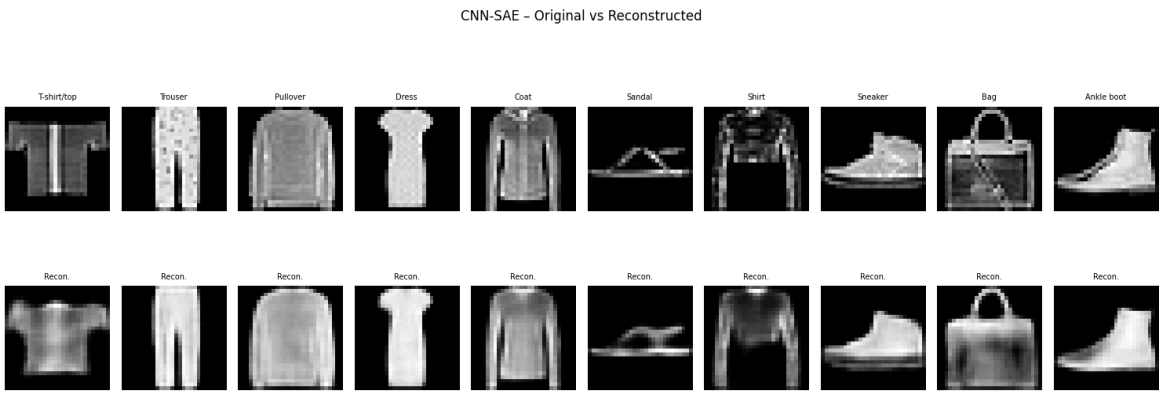
`pca_explained_variance.png`



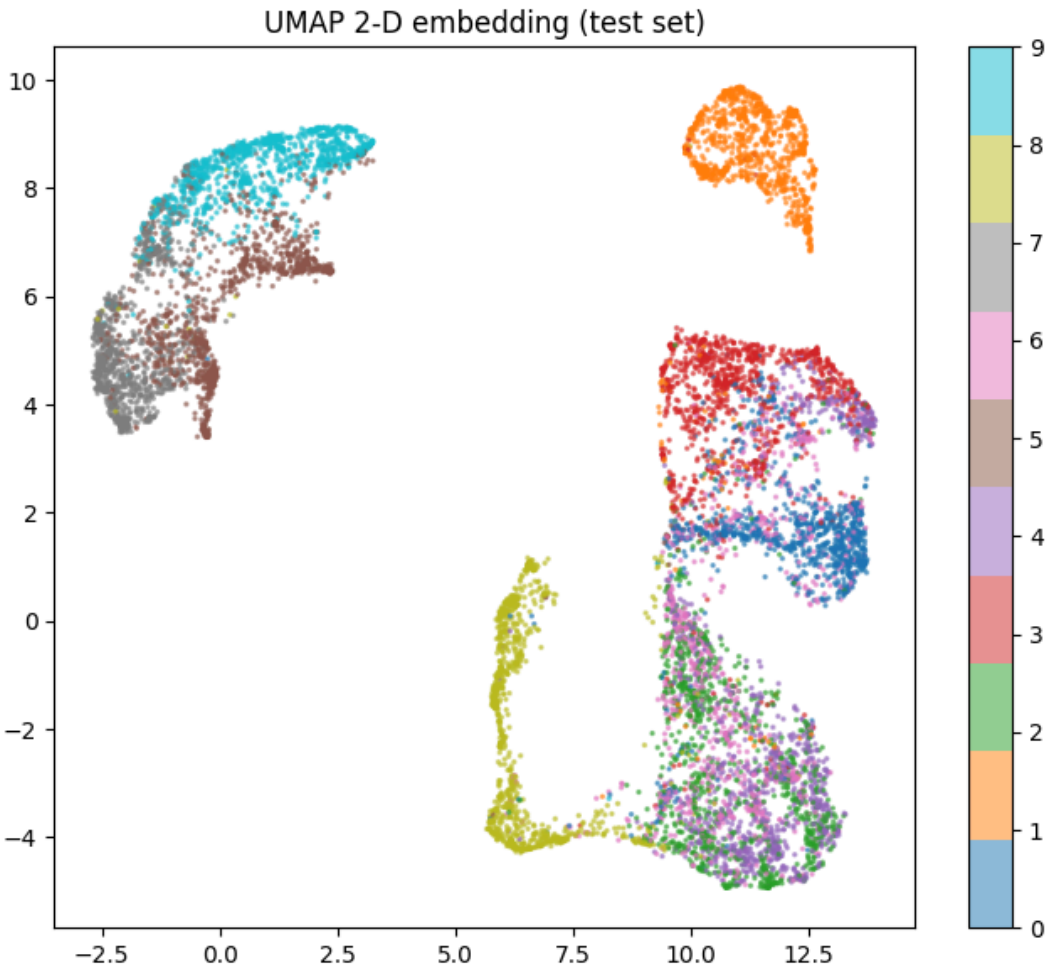
`sae_reconstructions.png`



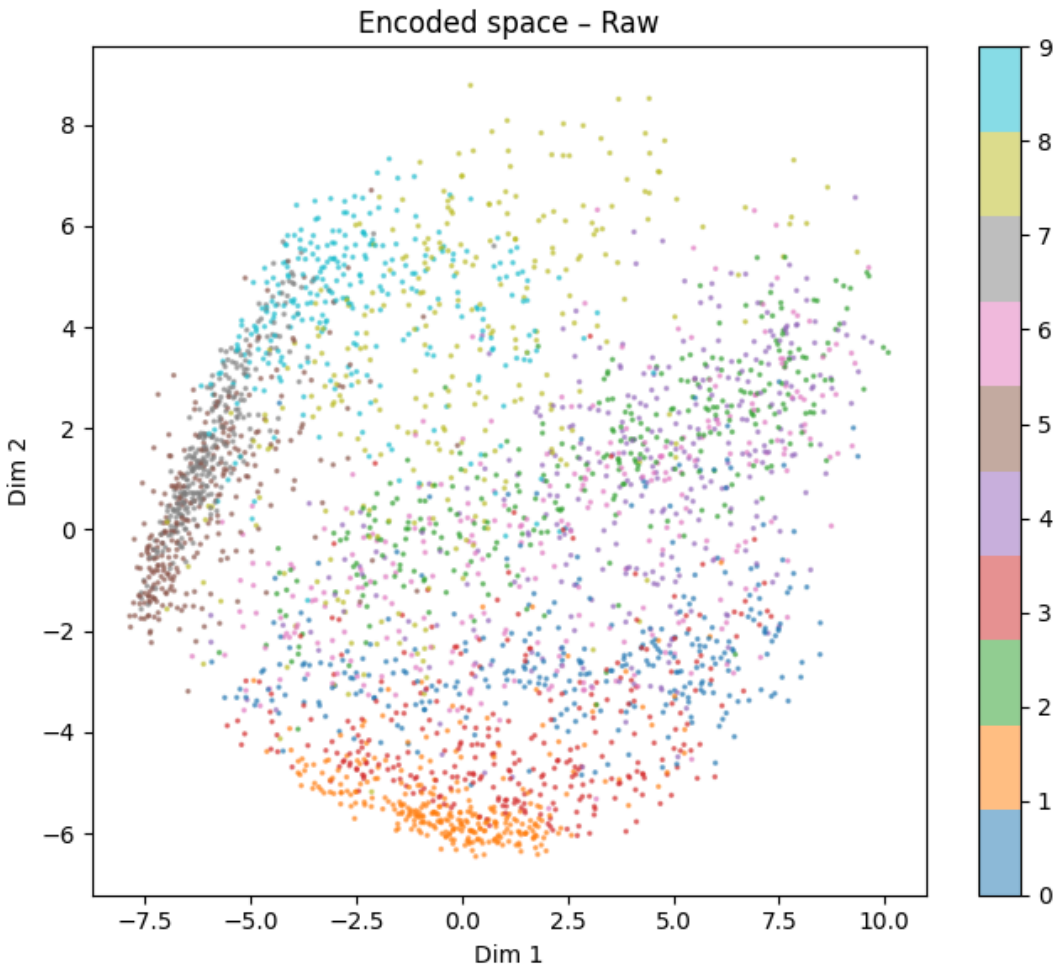
`cnnsae_reconstructions.png`



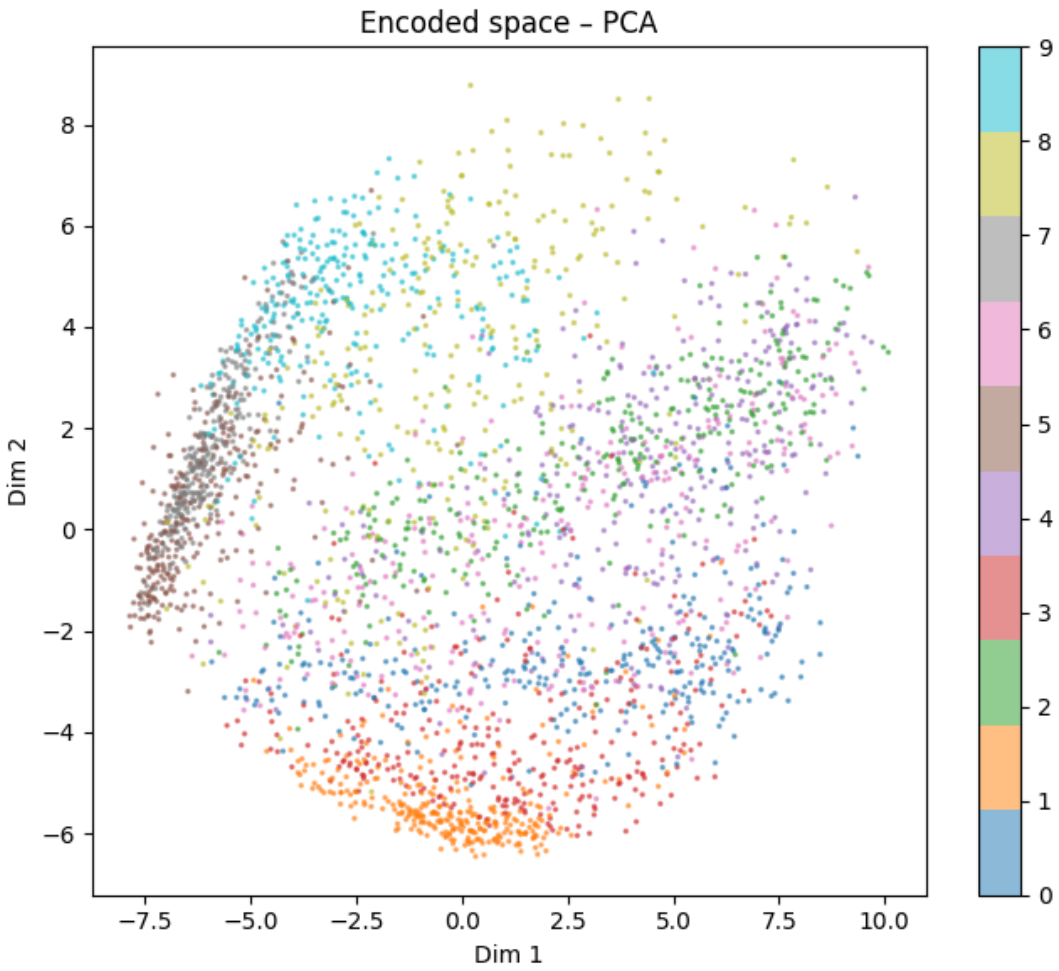
umap_2d.png



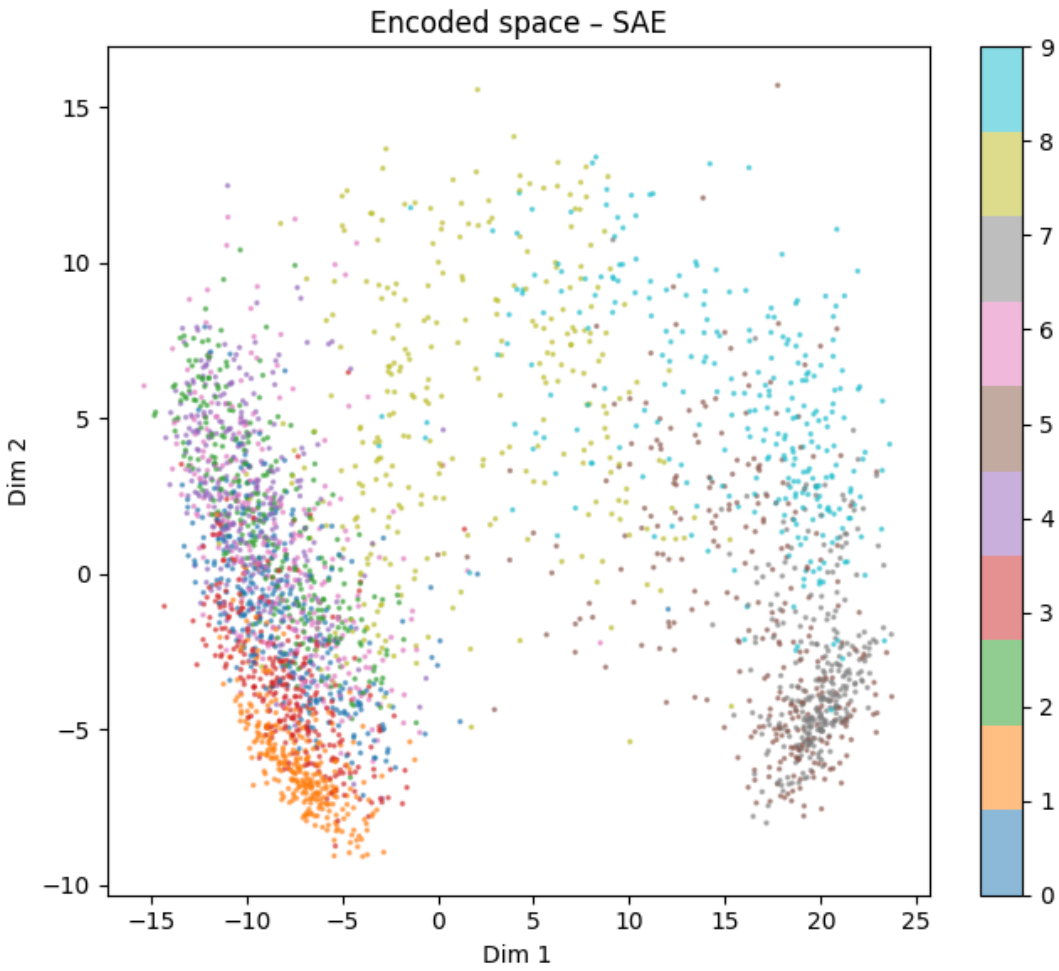
scatter_Raw.png



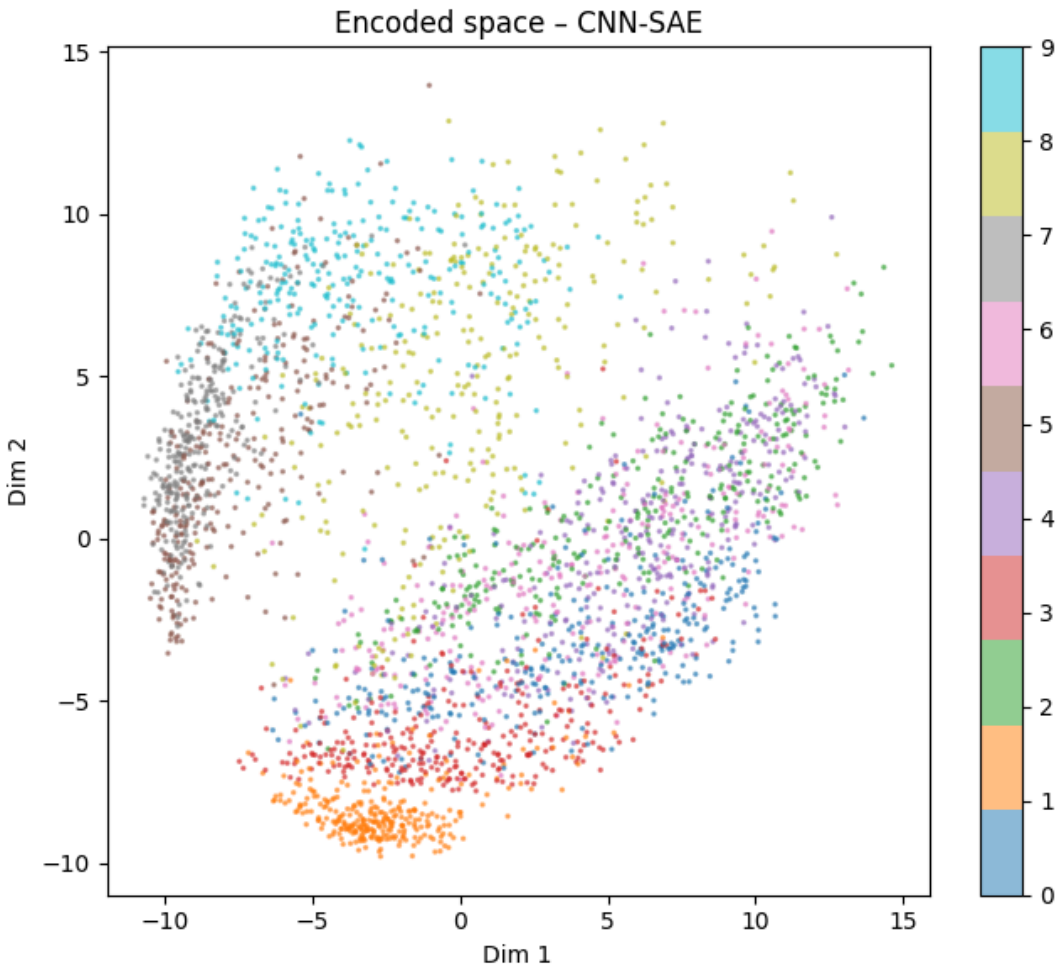
`scatter_PCA.png`



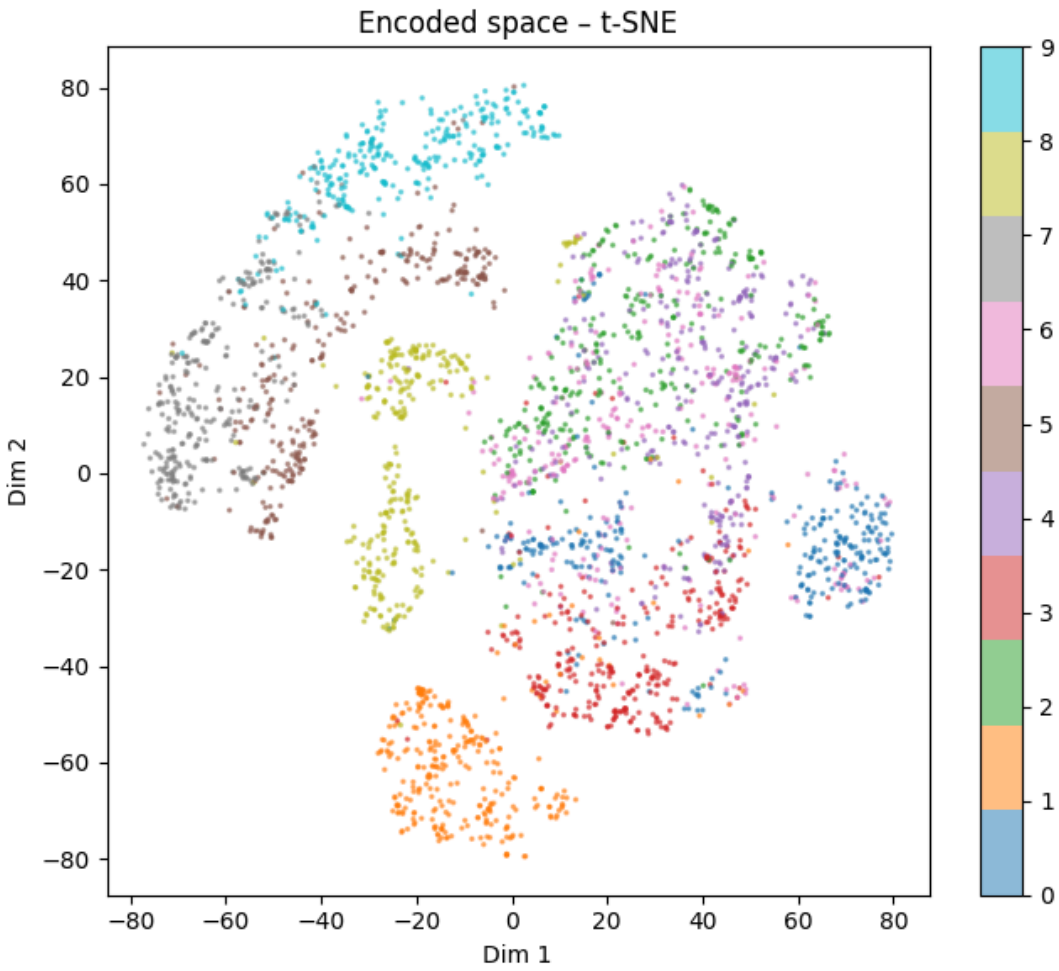
`scatter_SAE.png`



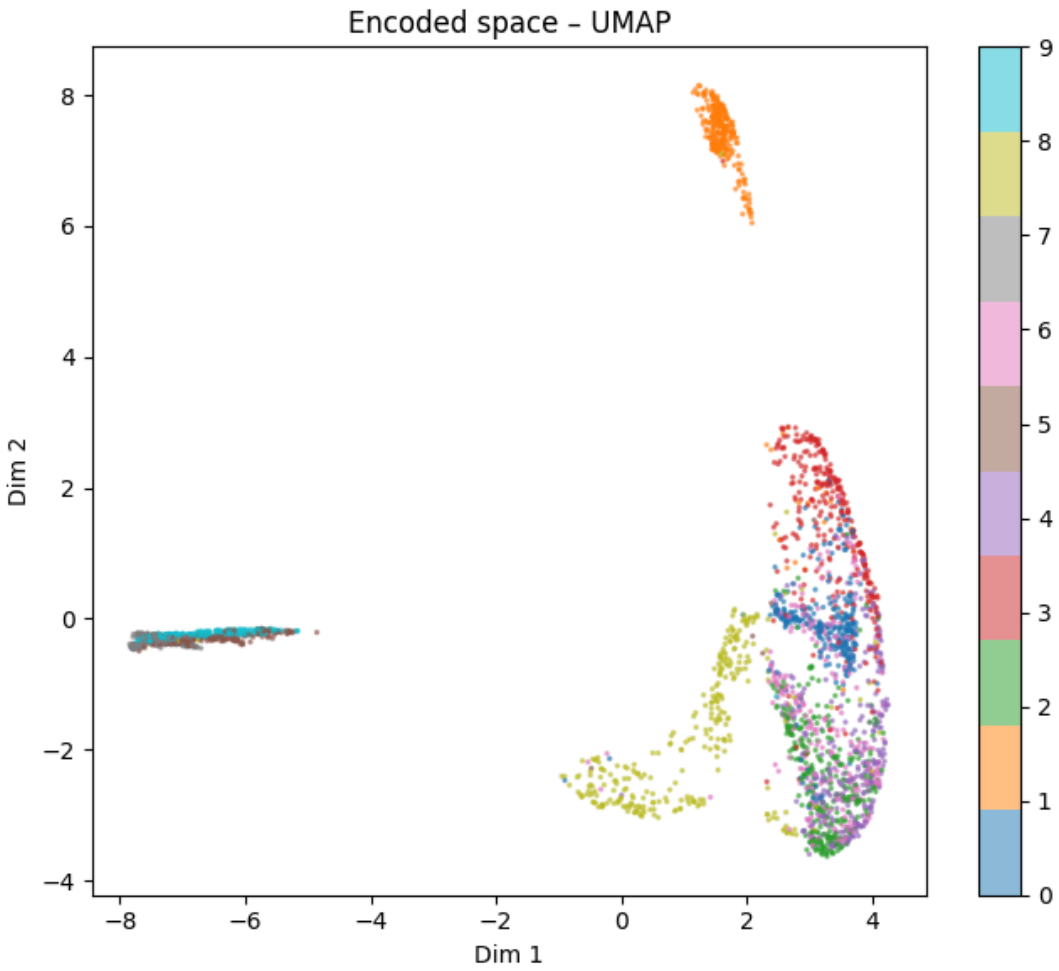
`scatter_CNN-SAE.png`



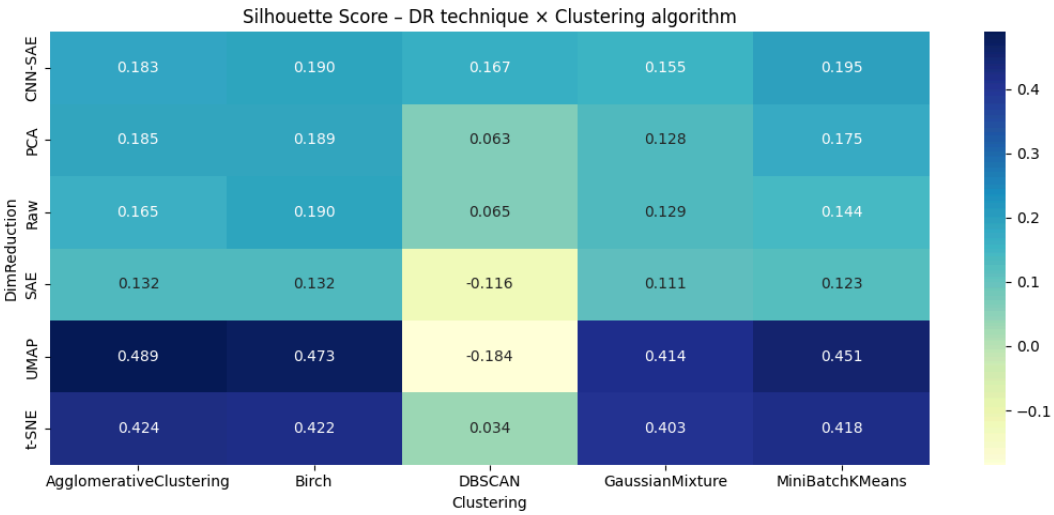
`scatter_t-SNE.png`



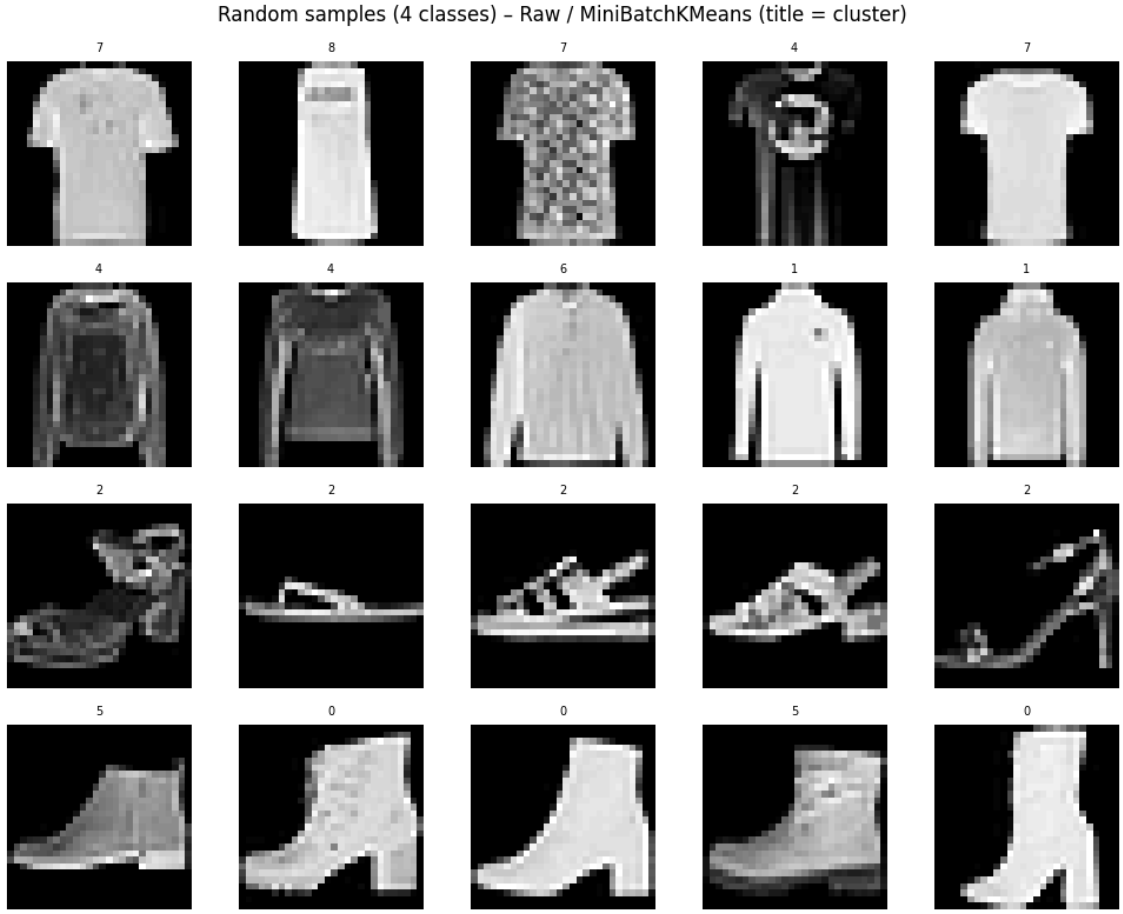
`scatter_UMAP.png`



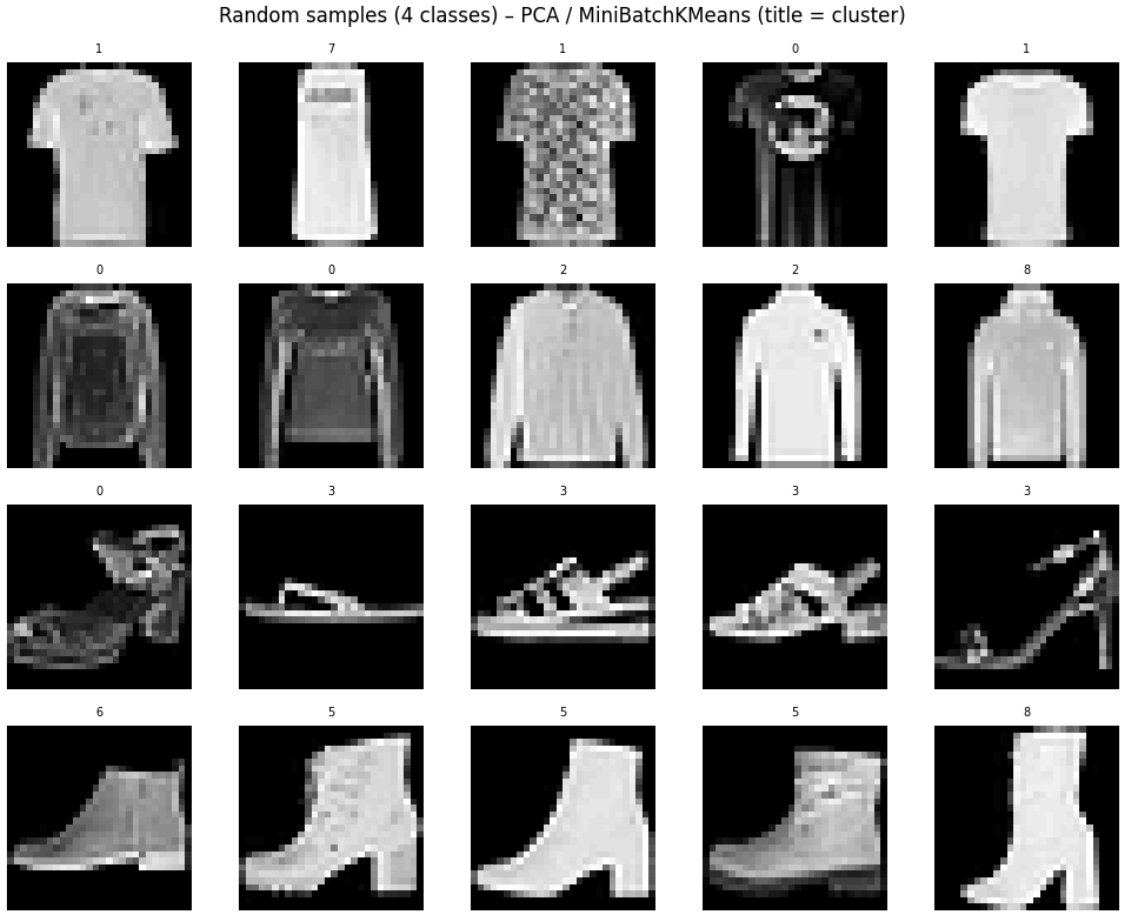
heatmap_silhouette.png



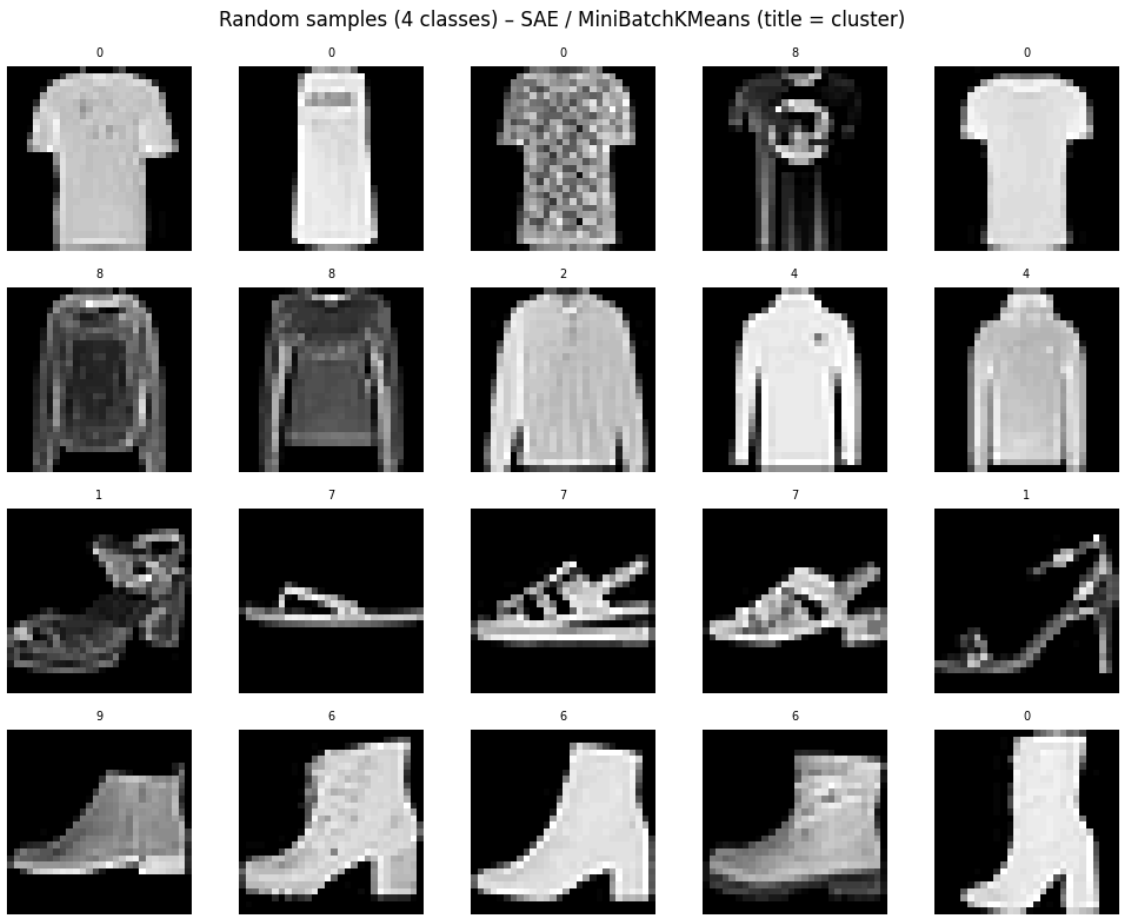
cluster_examples_Raw.png



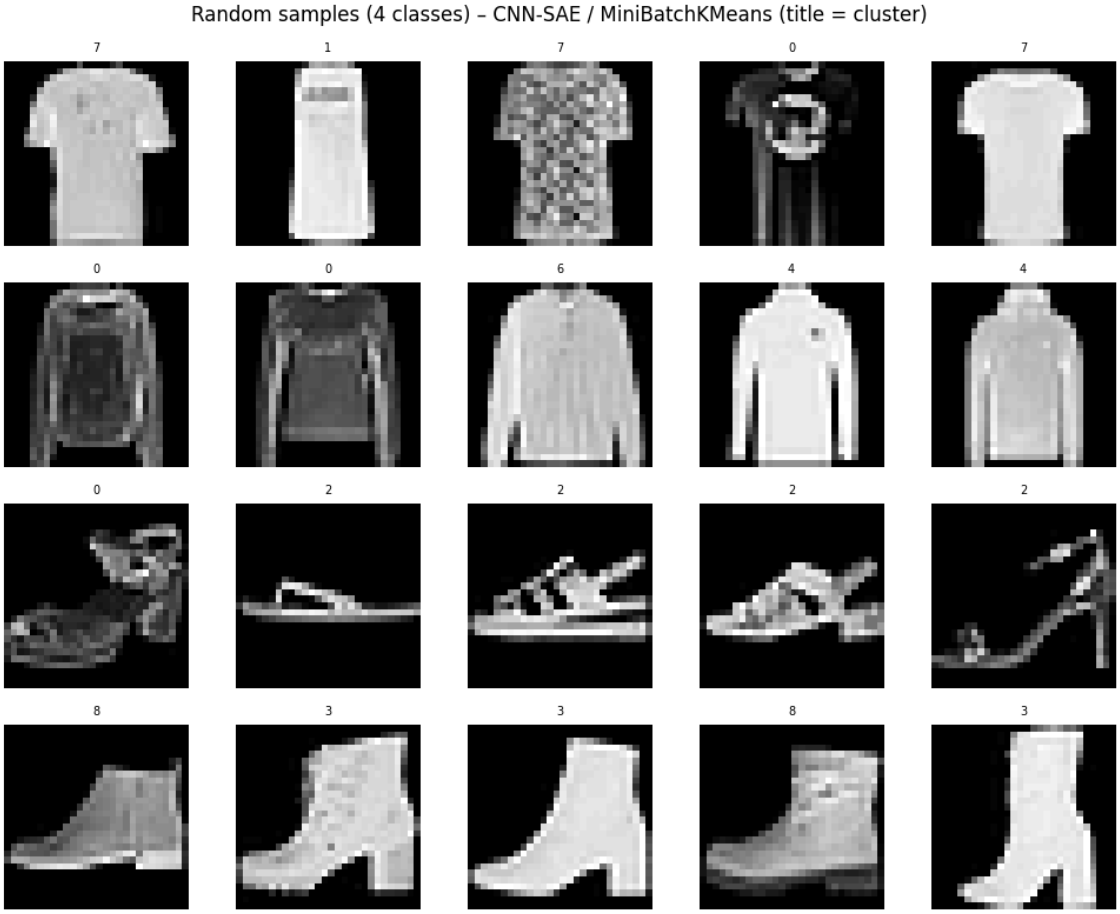
`cluster_examples_PCA.png`



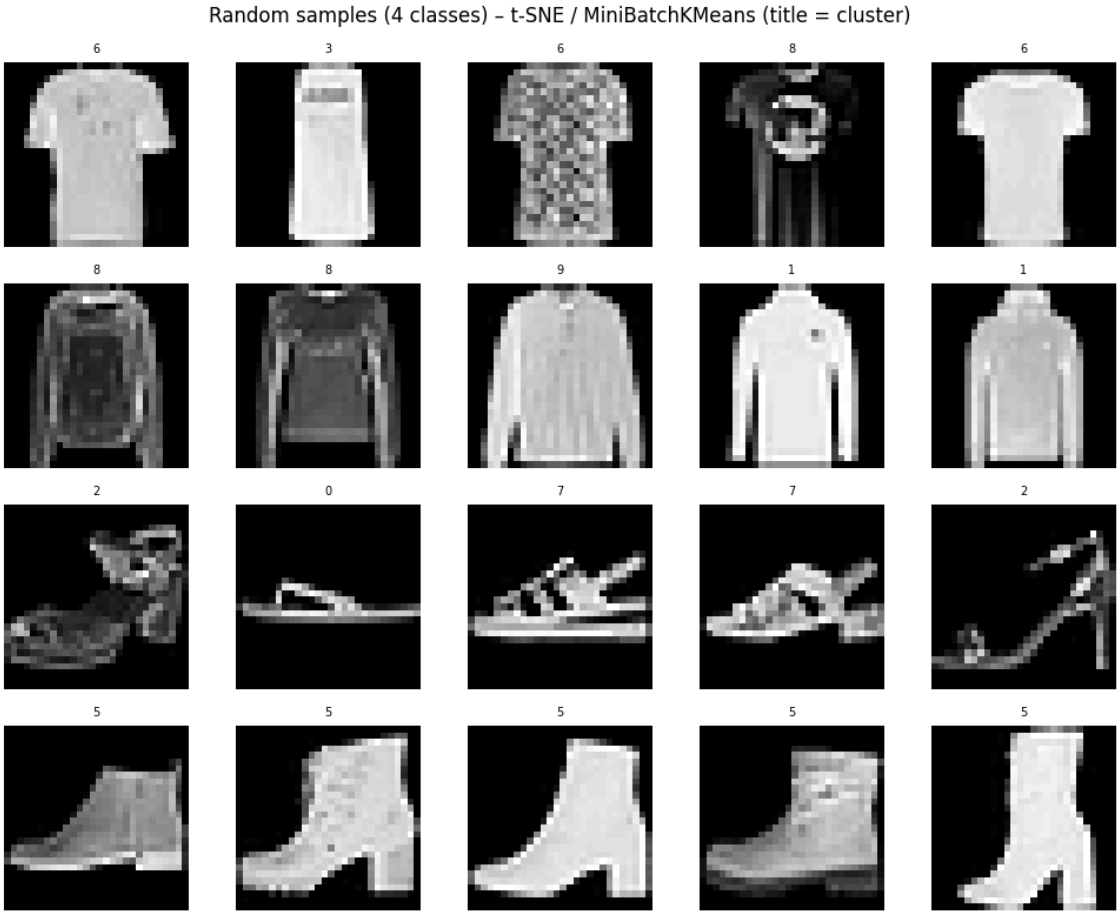
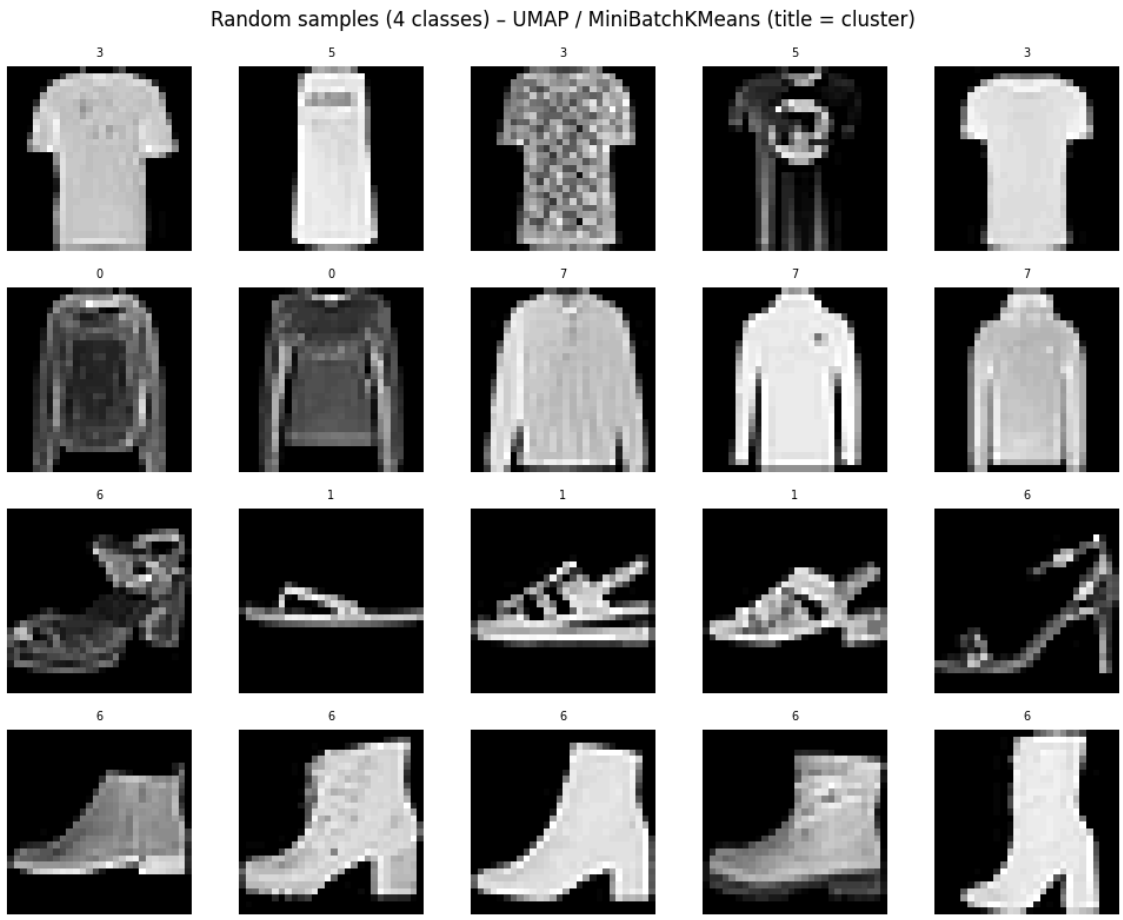
`cluster_examples_SAE.png`



`cluster_examples_CNN-SAE.png`



`cluster_examples_t-SNE.png`



`cluster_examples_UMAP.png`